

TERASIN TABLET

Ingredient(s):

Terasin Tablet 2mg:

Each tablet contains:

Terazosin 2mg

Terasin Tablet 5mg:

Each tablet contains:

Terazosin 5mg

Pharmacology (Summary of Pharmacodynamics and Pharmacokinetics):

Pharmacodynamics

Benign Prostatic Hyperplasia (BPH):

The symptoms associated with BPH are related to bladder outlet obstruction. An increase in smooth muscle tone in the prostate and bladder neck leads to constriction of the bladder outlet. Smooth muscle tone is mediated by sympathetic nervous stimulation of α_1 -adrenoceptors, which are abundant in the prostate, prostatic capsule and bladder neck. The reduction in symptoms and improvement in urine flow rates following administration of Terazosin is related to relaxation of smooth muscle produced by blockage of α_1 -adrenoceptors in the bladder neck and prostate. Terazosin is able to reduce the bladder outlet obstruction without affecting bladder contractility because relatively few α_1 -adrenoceptors are present in the bladder body.

Hypertension:

The vasodilatory hypotensive action of Terazosin appears to be produced mainly by blockade of α_1 -adrenoceptors. It decreases blood pressure gradually within 15 minutes following oral administration.

Although blockade of α_1 -adrenoceptors also lowers blood pressure in hypertensive patients with increased peripheral resistance, Terazosin treatment of normotensive patients with BPH did not result in a clinically significant blood pressure-lowering effect. There was a tendency for patients to gain weight during terazosin therapy. Decreases in hematocrit and total protein have been observed with α -blockade and are attributed to hemodilution.

Pharmacokinetics

Terazosin HCl administered as Terazosin tablet is essentially completely absorbed in man. The bioavailability of Terazosin is approximately 90%. Food had little or no effect on the extent of absorption but food delayed the time to peak concentration by about 1 hour. Terazosin has been shown to undergo minimal hepatic first pass metabolism and nearly all of the circulating dose is in the form of parent drug. The plasma levels peak about 1 hour after dosing. The half life of Terazosin is approximately 12 hours. A study that evaluated the effect of age on Terazosin pharmacokinetics showed that the mean half-lives were 14.0 and 11.4 hours for the age group ≥ 70 years and the age group of 20-39 years, respectively. After oral administration, the plasma clearance was decreased by 31.7% in patients ≥ 70 years compared to that in patients 20-39 years.

Terazosin is highly bound to plasma protein and binding is constant over the clinically observed concentration range. Approximately 10% of an orally administered dose is excreted as parent drug in the urine and approximately 20% is excreted in the feces. The remainder is eliminated as metabolites. Impaired renal function had no significant effect on the elimination of Terazosin, and dosage adjustment of Terazosin to compensate for the drug removal during hemodialysis (approximately 10%) does not appear to be necessary. Overall, approximately 40% of the administered dose is excreted in the urine and approximately 60% in the feces.

Indication(s):

Treatment of symptomatic benign prostatic hyperplasia (BPH).

Terazosin is also indicated for the treatment of hypertension. It can be used alone or in combination with other antihypertensive agents, eg diuretics or β -adrenergic blocking agents.

Dosage and Administration:

The dose of Terazosin hydrochloride should be adjusted according to the patient's individual response. The following is a guide to its administration:

Initial Dose

The recommended starting dose for all patients is 1 mg at bedtime. This dose should not be exceeded. This initial dosing regimen should be strictly observed to minimize the potential for severe hypotensive effects.

Subsequent Doses

Hypertension

The dose may be slowly increased to achieve the desired blood pressure response. The usual recommended dose range is 1 to 5mg administered once a day; however, some patients may benefit from doses as high as 20mg per day. Doses over 20mg do not appear to provide further blood pressure effect and doses over 40mg have not been studied. Blood pressure should be monitored at the end of the dosing interval to be sure control is maintained throughout the interval. It may also be helpful to measure blood pressure 2 to 3 hours after dosing to see if the maximum and minimum responses are similar, and to evaluate symptoms such as dizziness or palpitations which can result from excessive hypotensive response. If response is substantially diminished at 24 hours, an increased dose or use of a twice daily regimen can be considered. If Terazosin hydrochloride administration is discontinued for several days or longer, therapy should be reinstituted using the initial dosing regimen. In clinical trials, except for the initial dose, the dose was given in the morning.

Benign Prostatic Hyperplasia (BPH)

The dose may be slowly increased to achieve the desired clinical response in BPH patients. The usual recommended dose range is 5 to 10 mg administered once a day. Urine flow rate measured approximately 24 hours after the last dose has shown the beneficial effects in BPH persists for the recommended dosing interval. Symptom improvements have been detected as early as two weeks after starting treatment with Terazosin hydrochloride. Improvements in the flow rate may be seen somewhat later. If Terazosin hydrochloride administration is discontinued for several days or longer, therapy should be reinstituted using the initial dosing regimen.

Route of administration:

Oral administration.

Contraindication(s):

Patients known to be hypersensitive to Terazosin hydrochloride.

Precaution(s) / Warning(s):

Warnings

Syncope and "First-dose" Effect: Terazosin can cause marked lowering of blood pressure, especially postural hypotension, and syncope in association with the 1st dose of the first few days of therapy. A similar effect can be anticipated if therapy is interrupted for several days and then re-started. Syncope is associated with rapid dosage increases or the introduction of another hypertensive drug. Syncope is believed to be due to an excessive postural hypotensive effect, although the syncopal episode has been preceded by a bout of severe supraventricular tachycardia with heart rates of 120-160 beats/min.

To decrease the likelihood of syncope or excessive hypotension, the starting dose should always be initiated with a 1mg dose of Terazosin tablet, given at bedtime. The 2, 5, 10mg doses of Terazosin are not indicated for initial therapy. Dosage should then be increased slowly and additional antihypertensive agents should be added with caution. The patient should avoid situations, eg. driving or hazardous tasks, where injury could result should syncope occur during initiation of therapy.

If syncope occurs, the patient should be placed in a recumbent position and treated supportively as necessary.

Precautions

General: Prostatic Cancer: Carcinoma of the prostate and BPH causes many of the same symptoms. Since these 2 diseases frequently coexist, patients thought to have BPH should be examined prior to starting Terazosin therapy to rule out the presence of carcinoma of the prostate.

Orthostatic Hypotension: Syncope is the most severe orthostatic effect of Terazosin tablets. Other symptoms of lowered blood pressure, eg dizziness, lightheadedness and palpitations, were more common and occurred in some 28% of patients in clinical trials of hypertension. In BPH clinical trials, 21% of the patients experienced one or more of the following: Dizziness, hypotension, postural hypotension, syncope and vertigo. Caution should be exercised in treating patients with occupations in which such events represent potential problems.

Information for Patients: Patients should be informed of the possibility of syncopal and orthostatic symptoms, especially at the initiation of terazosin therapy, and to avoid driving or hazardous tasks for 12 hours after the first dose, after a dosage increase and after interruption of therapy when treatment is resumed. They should be cautioned to avoid a situation where injury could result should syncope occur during initiation of terazosin therapy. They should be advised to sit or lie down when symptoms of lowered blood pressure occur, and to be careful when rising from a sitting or lying position. If dizziness, lightheadedness or palpitations are bothersome, they should be reported to the physician so that dose adjustment can be considered.

Effects on the Ability to Drive or Operate Machinery: Drowsiness or somnolence can occur with Terazosin tablets, caution is, therefore, required in patients who must drive or operate heavy machinery.

Safety and effectiveness in children have not been determined.

Interaction with Other Medicaments

1. In patients receiving terazosin plus ACE inhibitors or diuretics the proportion reporting dizziness or related side effects was greater than in the total population of terazosin treated patients.
2. Caution should be observed when terazosin is administered with other antihypertensive agents to avoid the possibility of significant hypotension.
3. When adding terazosin to a diuretic or other antihypertensive agent, dosage reduction and retitration may be necessary.
4. Concomitant use of phosphodiesterase-5-inhibitors (e.g. sildenafil, tadalafil, vardenafil) and terazosin may lead to symptomatic hypotension in some patients.

Pregnancy and Lactation:

Teratogenic Effects: Pregnancy Category C: There are no adequate and well-controlled studies in pregnant women and the safety of terazosin in pregnancy has not been established. Terazosin is not recommended during pregnancy unless the potential benefit justifies the risk to the mother and fetus.

Use in lactation:

It is not known whether terazosin is excreted in breast milk. Because many drugs are excreted in breast milk, caution should be exercised when terazosin tablets are administered to a nursing woman.

Side Effect(s) / Adverse Reaction(s):

Terazosin, in common with other alpha-adrenoceptor blockers, may cause syncope. Syncope may occur within 30-90 minutes of the initial dose of the drug, and may occasionally occur in association with rapid dosage increases or the introduction of other antihypertensive agents.

Other adverse drug effects reported with terazosin are:

1. Blood and lymphatic system disorders: Thrombocytopenia
2. Immune system disorders: Anaphylactoid reaction
3. Psychiatric disorders: Depression, nervousness, anxiety, insomnia
4. Nervous system disorders: Dizziness, somnolence, headache, paraesthesia, vertigo
5. Eye disorders: Blurred vision, amblyopia, visual impairment, conjunctivitis
6. Ear and labyrinth disorders: Tinnitus
7. Cardiac disorders: Palpitations, tachycardia, arrhythmia, atrial fibrillation
8. Vascular disorders: Postural hypotension, syncope, vasodilatation
9. Respiratory, thoracic and mediastinal disorder: Nasal congestion, rhinitis, dyspnoea, sinusitis, epistaxis, flu symptoms, pharyngitis, cold symptoms, cough
10. Gastrointestinal system disorders: Nausea, abdominal pain, constipation, diarrhoea, dry mouth, dyspepsia, flatulence, vomiting
11. Skin and subcutaneous tissue disorders: Pruritus, rash, hyperhidrosis, angioedema
12. Musculoskeletal and connective tissue disorders: Back pain, pain in extremity, neck pain, shoulder pain, gout, arthralgia, arthritis, joint disorders, myalgia
13. Renal and urinary disorders: Pollakiuria, urinary tract infection and urinary incontinence (primarily reported in post-menopausal women)
14. Reproductive system and breast disorders: Libido decreased, erectile dysfunction, priapism
15. General disorders and administration site conditions: Asthenia, peripheral oedema, oedema, chest pain, face oedema, pyrexia
16. Investigations: Weight increased. Decreased haematocrit, decreased haemoglobin, decreased white blood cell count, decreased total protein and decreased blood albumin (suggestive of haemodilution). Treatment with terazosin for up to 24 months had no significant effect on prostate specific antigen (PSA) levels.

Symptoms and Treatment for Overdosage, and Antidote(s):

Should overdosage of terazosin lead to hypotension, support of the cardiovascular system is of first importance. Restoration of blood pressure and normalization of heart rate may be accomplished by keeping the patient in the supine position. If this measure is inadequate, shock should first be treated with volume expanders. If necessary, vasopressors should then be used and renal function should be monitored and supported as needed. As laboratory data show that terazosin is highly protein-bound; dialysis may not be of benefit.

Storage Condition(s):

Store at temperature below 30°C. Protect from light and moisture.

Shelf-Life:

3 years from the date of manufacture.

Product Description and Packing(s):

Terasin Tablet 2mg:

A round yellow tablet in the shape of " " " "

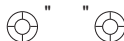
7's x 4, 10's x 10 and 10's x 50.



Terasin Tablet 5mg:

A brownish red tablet in the shape of " " " "

7's x 4, 10's x 10 and 10's x 50.



Manufacturer and Product Registration Holder:
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